



Where Do We Stand on Anti-Ku Antibodies? Prevalence and Clinical Impact in Light of Recent Findings

A.NASSAR(1), S.BOUAFIA(2), S. ZAGHES (2), Y. REHABI (2), A. ALLIOUCH KERBOUA(2), H. MERICHE (2), N. S. GADIRI(2).

(1) Didouche Mourad Hospital, Constantine, Algeria

(2) Immunology Department, Sainte Thérèse Clinic, University Hospital Center (CHU) Annaba, Annaba, Algeria

Introduction

Anti-KU antibodies belong to a category of autoantibodies directed against a nuclear protein complex involved in DNA repair and cell cycle regulation. Although they are considered rare, they are likely underdiagnosed due to the scarcity of specific tests and a lack of clinical awareness. These autoantibodies are detected across a broad spectrum of autoimmune diseases, including various connective tissue diseases, but do not correspond to a single or specific connective tissue pathology. Their presence has been reported in overlap syndromes with diverse clinical manifestations, ranging from systemic sclerosis to inflammatory myositis. However, their frequencies of occurrence and precise clinical correlations remain poorly characterized in certain populations, particularly in hospital settings and in multicenter cohorts. The main objective of this study is to evaluate the prevalence of anti-KU antibodies as well as the clinical associations observed in patients followed in different hospital departments, in order to better understand their diagnostic and prognostic role in autoimmune diseases.

Materials and methods

1/ Patients:

A descriptive retrospective study was conducted on 40 patients from a hospital registry between 2018 and 2024. Clinical, biological, and serological data were analyzed. Anti-KU antibody status was classified into three levels of positivity. Demographic variables, clinical syndromes, care departments, and age were examined. Prevalences and statistical distributions were assessed.

2/ Methods:

Patient serum analysis was performed using indirect immunofluorescence (IIF) on HEp-2 cells and by an immuno-dot technique.

Results

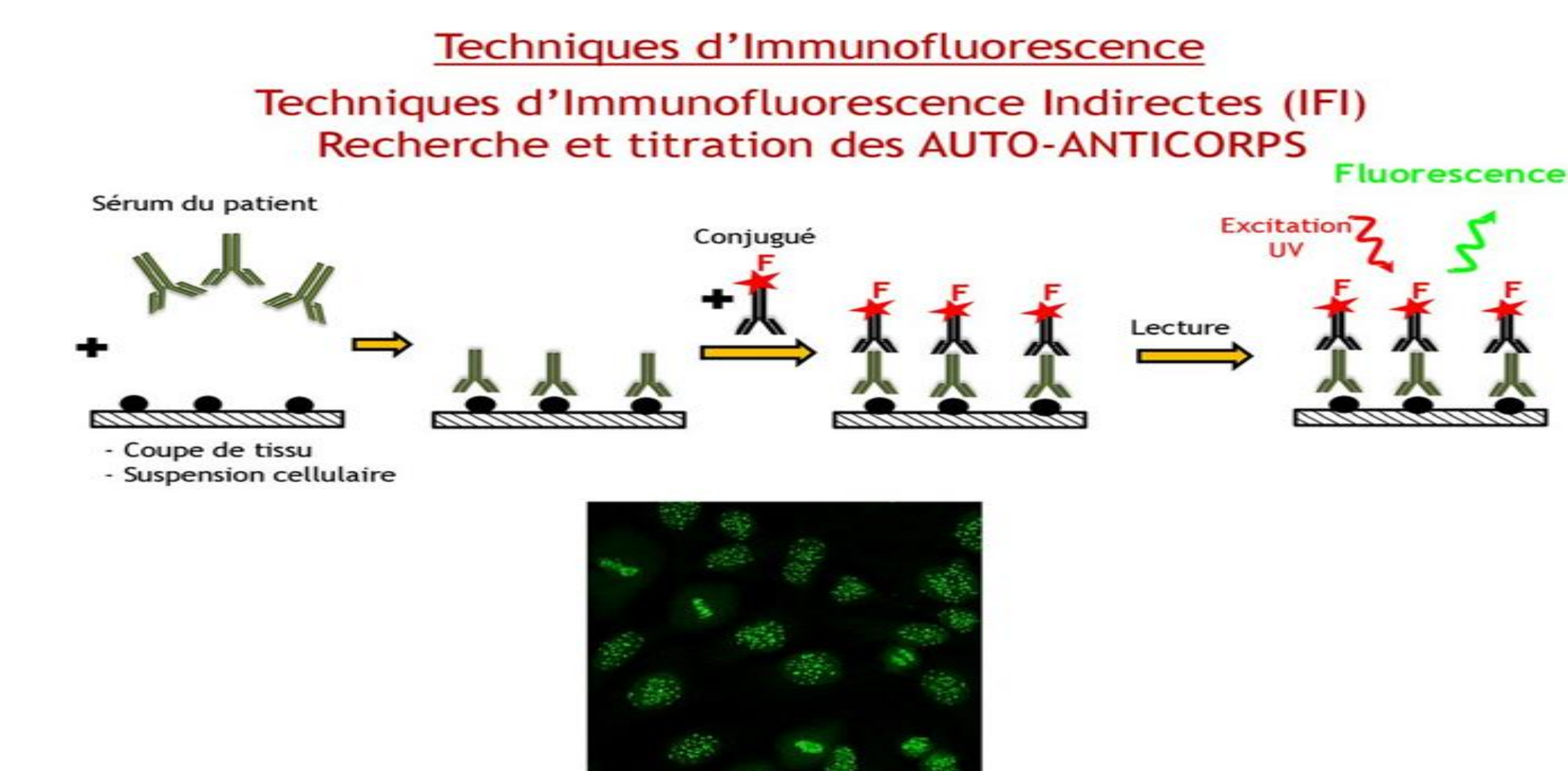


Figure 1. Principle of indirect immunofluorescence

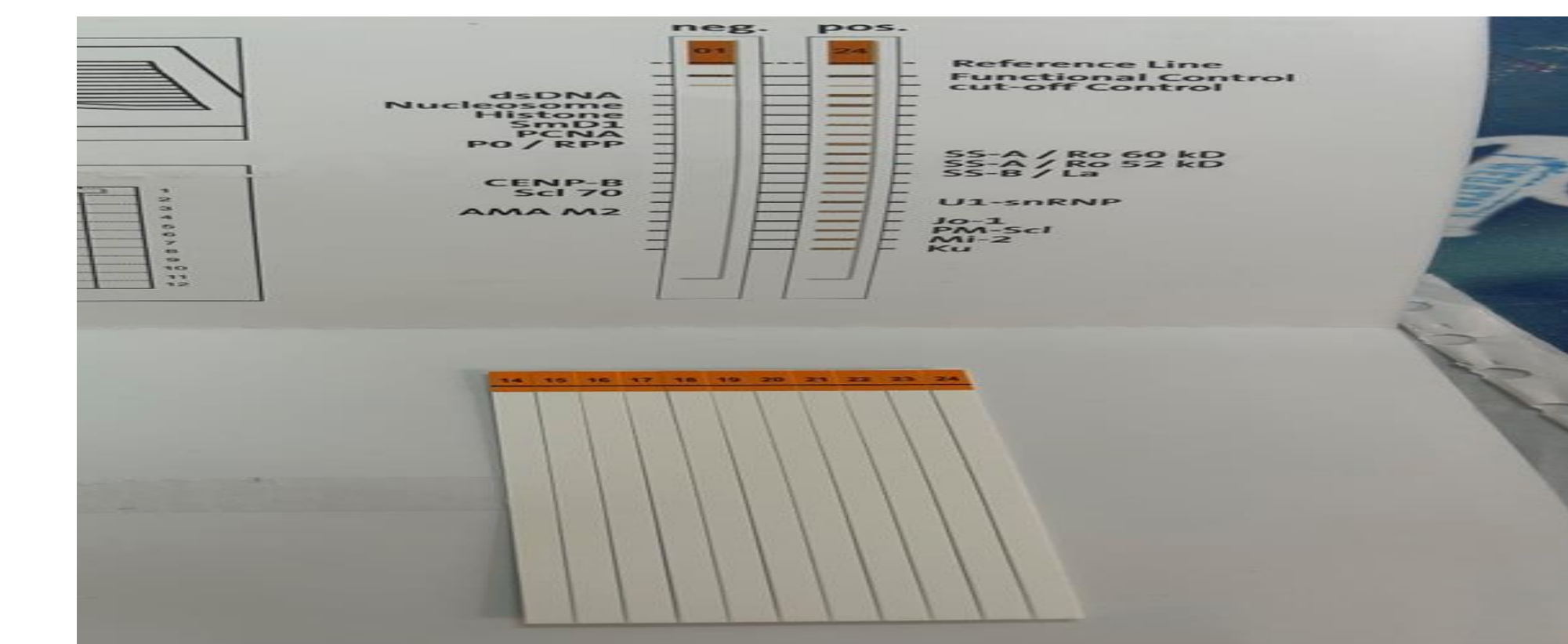


Figure 2. Immuno Dot

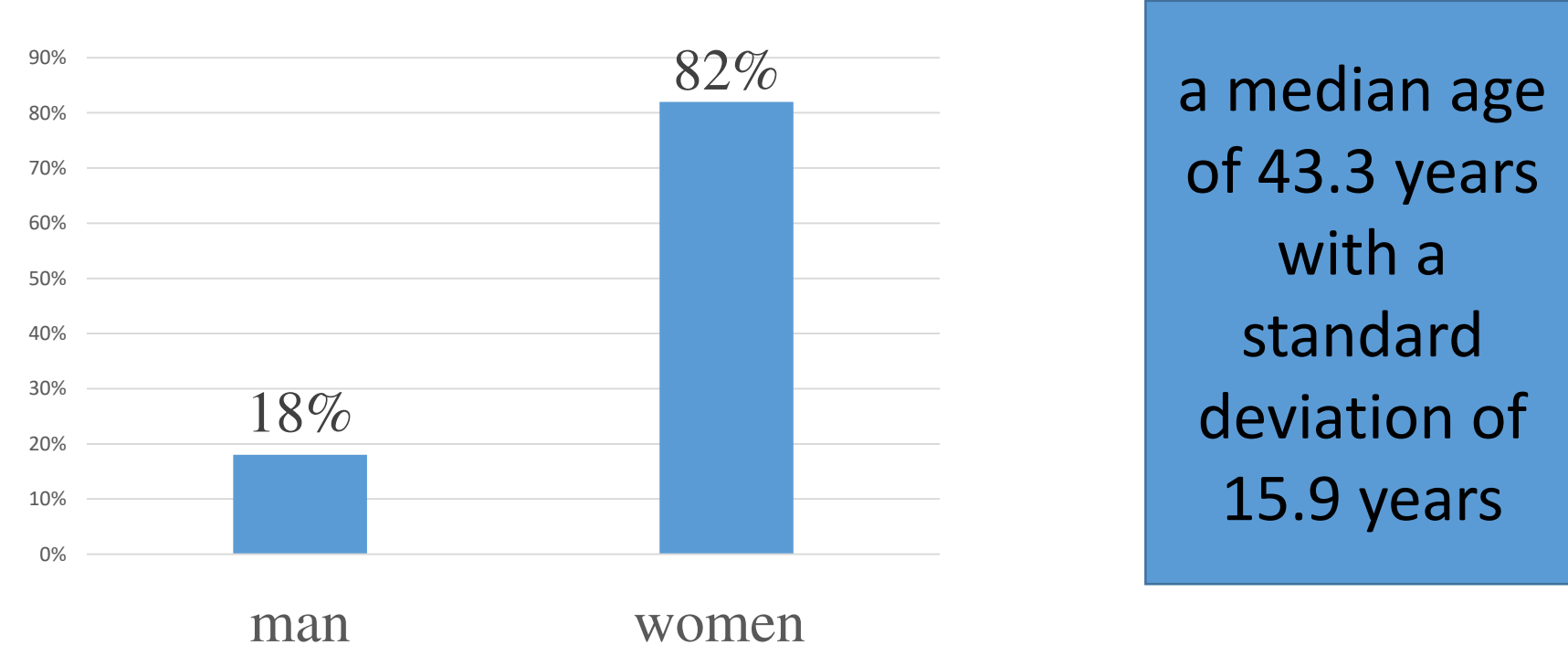


Figure 3. Epidemiological characteristics of our study group

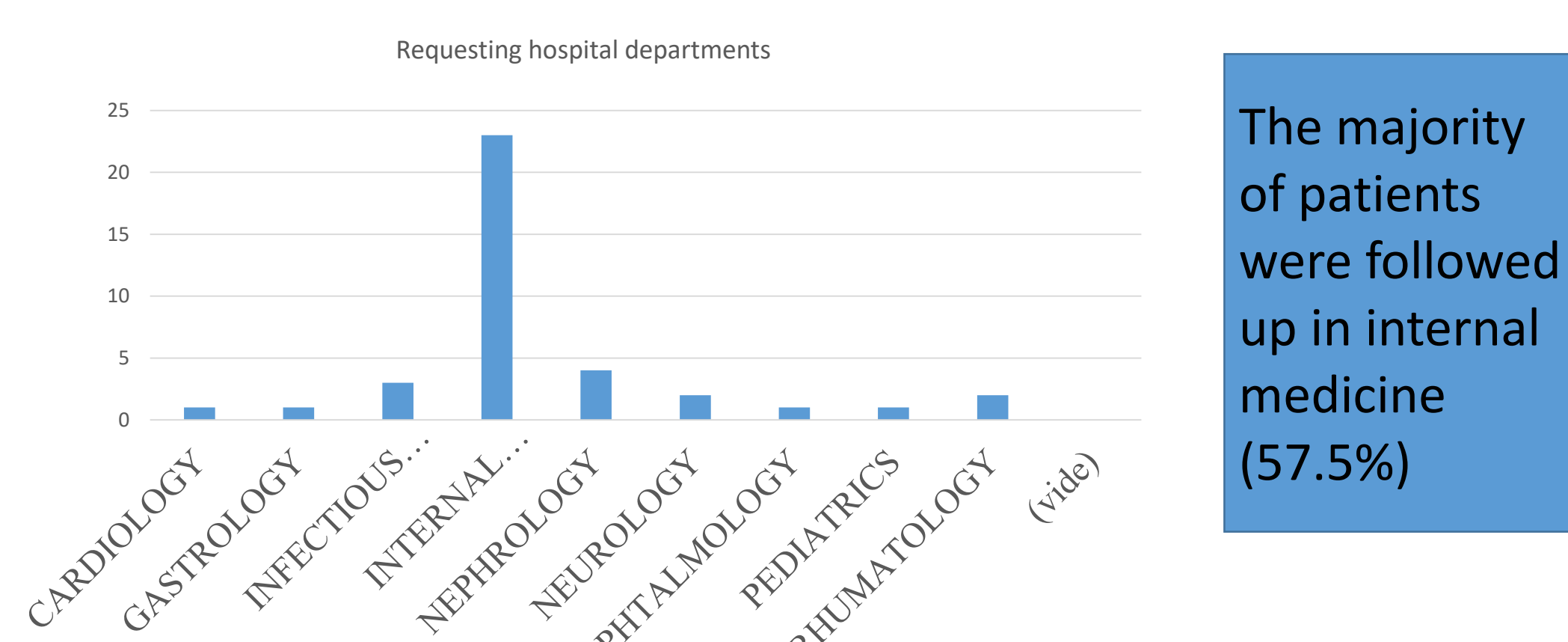


Figure 4. Requesting hospital departments

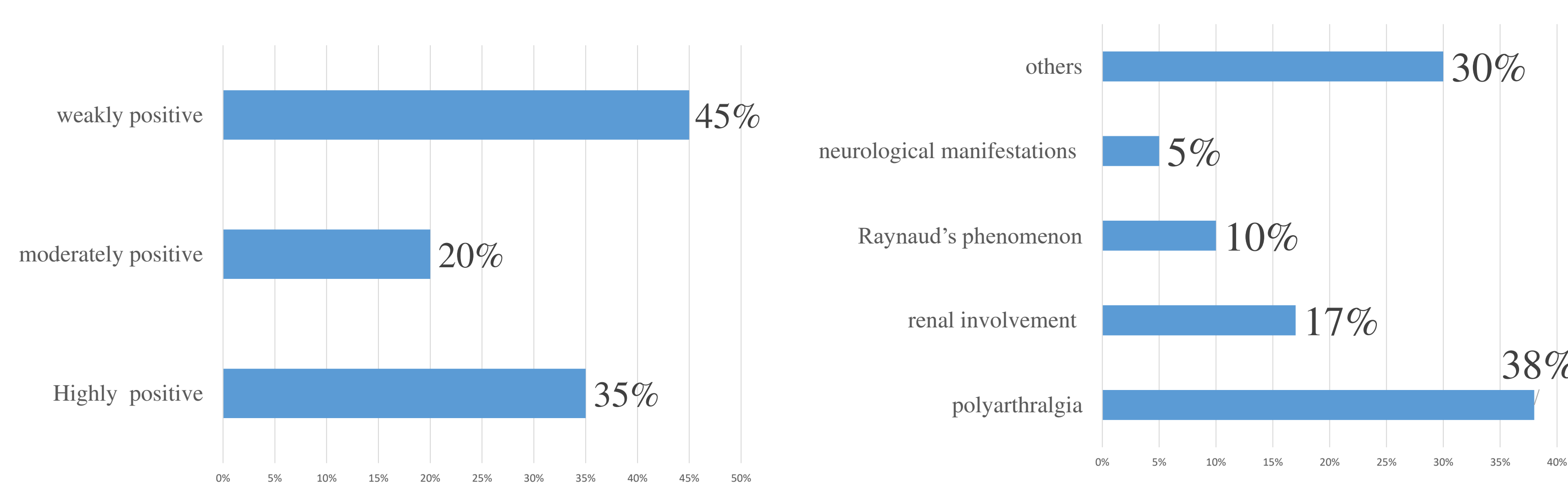


Figure 5. Graphical presentation of the study results on anti-Ku antibody

Figure 6. Clinical manifestations associated with anti-Ku antibodies

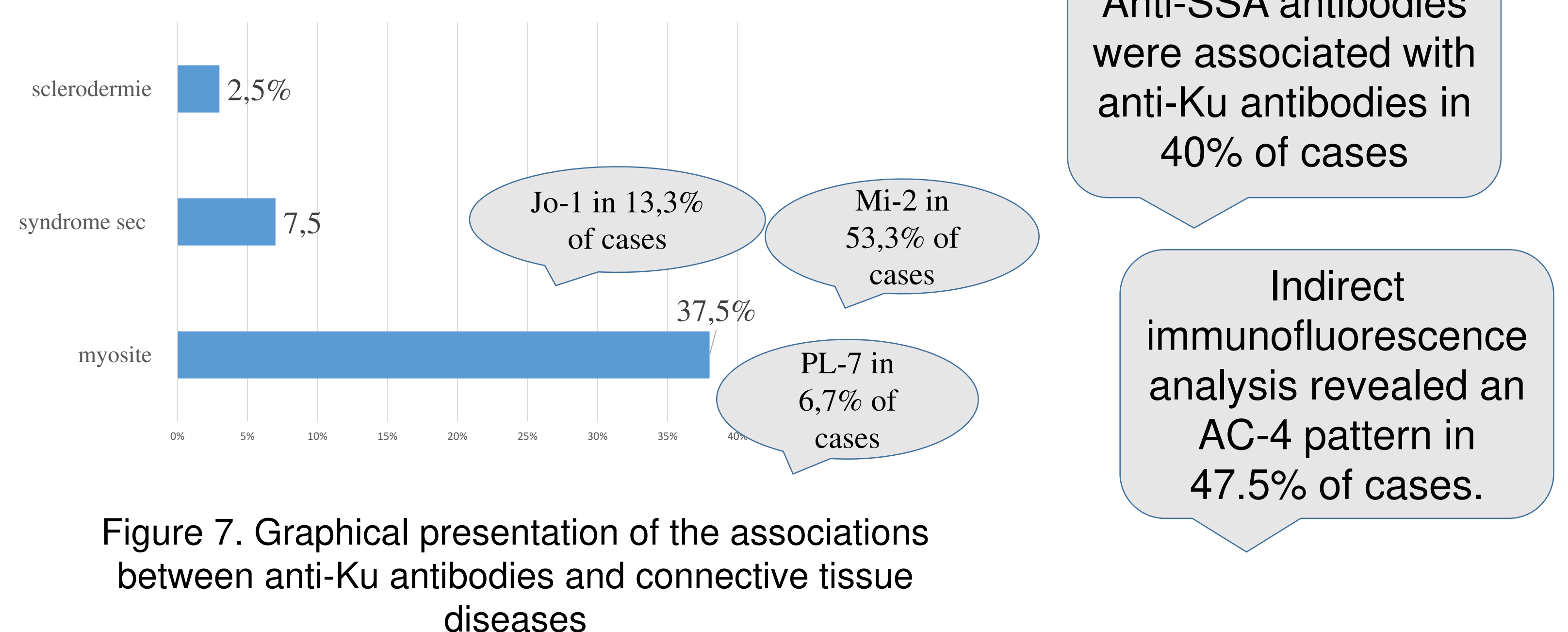


Figure 7. Graphical presentation of the associations between anti-Ku antibodies and connective tissue diseases

Discussion

Anti-Ku antibodies are rare autoantibodies, predominantly detected in middle-aged women, and associated with a broad spectrum of systemic autoimmune diseases. They are particularly characteristic of overlap syndromes, where manifestations of inflammatory myositis and systemic sclerosis coexist, but they are also found in other connective tissue diseases.

Patients with anti-Ku antibodies frequently present with autoimmune myositis, manifesting as proximal muscle weakness, myalgia, and elevated muscle enzymes such as creatine phosphokinase (CPK). Raynaud's phenomenon is also a common symptom. Other clinical manifestations include renal involvement, neurological manifestations—notably cranial nerve involvement—as well as arthralgia, arthritis, interstitial lung disease, atonic esophagus, and cardiac disorders such as myocarditis or arrhythmias.¹

Anti-Ku antibodies are often detected in association with other myositis-specific autoantibodies, such as anti-PL-7, anti-Jo-1, and anti-Mi-2, or with anti-SSA antibodies, underscoring the frequent autoimmune overlap in these patients.

Their presence is confirmed by indirect immunofluorescence on HEp-2 cells, where they produce a characteristic AC-4 pattern. These autoantibodies are considered non-specific but prognostic markers in systemic connective tissue diseases, particularly in inflammatory myositis and overlap syndromes. They often indicate an overlap phenotype including polymyositis–scleroderma.⁴

Our findings are consistent with current scientific data: several retrospective studies and clinical analyses report a low prevalence of anti-Ku antibodies (approximately 1 in 3,500 positive antinuclear antibody tests), a frequent association with overlap syndromes involving myositis, and a generally favorable prognosis with low progression to severe or malignant forms.

Conclusion

Anti-Ku antibodies are rare autoantibodies, predominantly observed in middle-aged women, and associated with a broad spectrum of autoimmune diseases, particularly myositis and overlap syndromes. They exhibit marked clinical heterogeneity, with frequent manifestations such as polyarthralgia, renal involvement, and Raynaud's phenomenon. Often co-present with other autoantibodies (anti-Mi-2, anti-Jo-1, anti-SSA), anti-Ku antibodies appear as a non-specific but potentially prognostic marker of systemic connective tissue disease.

References

Anti-Ku antibodies: what clinical picture?

Bimodal cardiac involvement in scleromyositis with anti-Ku antibodies

Camptocormia with trigeminal involvement: anti-Ku myositis

Autoimmune myositis: from the laboratory to the clinic

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